Guarded quadratic trace elimination for Diophantine consistency

Research for Vladimir Reshetnikov and ProveIt
AI-assisted construction with independent finite checks

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1 Guarded quadratic trace elimination

Two new fixed ordinary polynomials have thirteen unrestricted integer unknowns:

$$(\deg, \mathrm{ops}) = (118, 114), \qquad (120, 112).$$

The charged operation counts (multiplication, addition, subtraction) are $(66, 27, 21)$ and $(65, 26, 21)$. Each point strictly dominates the incoming thirteen-integer $(152, 119)$ and $(156, 117)$ points in both varying metrics; the two new points themselves trade degree against cost. All six full fixed numeral payloads and their finite recipe are retained unchanged. These are ordinary reductions through an identified native/compiler/refutation theorem, not a consistency assertion, global optimum or new end-to-end kernel certificate.

1.1 Entire literal equations

The inputs, in order, are

$$j, C, z, h, r, v, w, T, c, n, y, p_2, p_3 \in \mathbb{Z}.$$

Use the authentic ports $\beta, K_0, M_C, M_F, A_c, B_c$, with M_F already shifted, $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Their full bytes, binding and finite recipe are under the retained coefficient directory:

`roots/inherited_round22/inherited_round20/`
`inherited/coefficients/`

All following names are polynomial abbreviations, not further unknowns:

$$J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \quad X = qW, \quad Y = v,$$
$$F = (K_0 + W)C - zq_m, \qquad R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J,$$
$$E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = XY^2,$$
$$\eta = c - Yk, \quad L = \ell k, \quad I = (8r + 4)c^2, \quad D_s = \Delta I^2, \quad F_s = D_s + 1,$$
$$Q = \Delta D_s, \quad \delta = Q - 1, \quad b = cT - R, \quad d_y = \delta y, \quad t = c + d_y, \quad t_t = t^2, \quad L_b = F_s b^2,$$
$$A_r = L_b + t_t - Qd_y y - 1, \qquad H_r = A_r^2 - 4L_b t_t,$$
$$q_3 = q^3, \quad M = 2q_3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2,$$
$$Z_0 = Un + MK_{\mathrm{lo}} + H(Y - q_3), \quad D_{\mathrm{err}} = (A_c F - q)^2 + (B_c C - q)^2.$$

The two charts are

$$k_{118} = R + 1 + h\ell, \qquad \mathcal{B}_{118} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big[4(Y\eta)^2 + (kq^2)^2\big], \qquad Z_{118} = Z_0 + UL,$$
$$k_{120} = R + 1 + hE, \qquad \mathcal{B}_{120} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big[4(Y\eta)^2 + (kq^3)^2\big], \qquad Z_{120} = Z_0.$$

In the 120-chart evaluator, $L = E(Yk)$ avoids a separate computation of ℓ. Finish in either chart with

$$G = \mathcal{B} - 1 - p_2^2 - p_3(p_3+1), \quad D_\tau = 1 + L(L+k), \quad U_2 = U^2, \quad g_a = U_2 G, \quad d_b = U_2 D_\tau,$$
$$N = (Z^2 - g_a - d_b)^2 - 4g_a d_b, \qquad P = NH_r + N + H_r = (N+1)(H_r+1) - 1.$$

The expanded final schedule saves one operation. There is no division gate, variable exponent, omitted numeral or hidden positivity restriction.

1.2 Resultant identity and the exceptional branch

For proof only, introduce f and put $V = fb - c$ and $E_a = V^2 - \delta y(2V + y)$. The complete identities are

$$A_r = F_s b^2 + c^2 - 1 - \delta y(y - 2c), \tag{1}$$
$$E_a - 1 = A_r - 2btf + b^2(f^2 - F_s), \tag{2}$$
$$H_r = A_r^2 - F_s(2bt)^2. \tag{3}$$

Thus $f^2 = F_s$ and $E_a = 1$ imply $H_r = 0$. The converse requires $bt \neq 0$; without it a resultant may vanish without a rational trace. We establish that nonvanishing from the guard *before* recovering f. Sharing t_t in the first and third expressions saves two evaluator gates; identity (1) must be used before assigning degrees.

1.3 All signed roots and guard restoration

Unconditionally q is even and nonzero, $|W| \geq 4$, $|X| \geq 8$, H is odd and nonzero, and $U \neq 0$. Since $a = (X+1)Y$ is zero or has $|a| \geq 7$, $\Delta = (a+2)^2 - 1 \geq 3$. Also ℓ is even and $D_\tau = 1 + \ell(\ell+1)k^2 \geq 1$.

Both N and H_r are squares modulo four. At $P = 0$, the integer factors $N+1, H_r+1$ are units, neither can be -1, and consequently $N = H_r = 0$. With $\rho = Z/U$,

$$(\rho^2 - G - D_\tau)^2 = 4GD_\tau.$$

This forces $G \geq 0$, including $G = 0$, and $\mathcal{B} > 0$, hence $Yk \neq 0$. Moreover

$$D_\tau \geq 1 + \tfrac{1}{2}X^2Y^4k^2 \geq 1 + 8q^2Y^4k^2 > q^2Y^2k^2 \geq \mathcal{B} > G.$$

Therefore $\rho \neq 0$ before any division by it. The rationals

$$p = \frac{\rho^2 + G - D_\tau}{2\rho}, \qquad \tau = \rho - p$$

satisfy $p^2 = G$ and $\tau^2 = D_\tau$. A rational whose square is integral is integral, by reducing its fraction to lowest terms. Thus $p, \tau \in \mathbb{Z}$. This handles zero guard roots, both signs and coincident square classes.

Set $\sigma = 1$ for degree 118 and $\sigma = 0$ for degree 120, and restore

$$p_{1,\mathrm{old}} = p, \qquad n_{\mathrm{old}} = n + \sigma L - (2L + k) - p + \tau.$$

With $Z_{\mathrm{old}} = Un_{\mathrm{old}} + MK_{\mathrm{lo}} + H(Y - q^3)$, one has $Z_{\mathrm{old}}/U + 2L + k = 2\tau$. Hence the identity

$$(Z_{\mathrm{old}} + Uk)^2 + 4U(LZ_{\mathrm{old}} - U) = (Z_{\mathrm{old}} + U(2L+k))^2 - 4U^2 D_\tau$$

restores the old carrier row, while $p^2 = G$ restores its guard.

1.4 Noncircular exclusion of every resultant escape

Only the positive budget and the retained centered-disk/authentic-mask argument are now used. Write $q_0 = |q|$ and $B_0 = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2$. The 118 guard is $Y^2B_0 - k^2q^6 > 0$, so $B_0 > 0$ and $|Y| > q_0^2$; the 120 guard similarly gives $|Y| > q_0^3$. The disk gives $D_{\mathrm{err}} < q^2$, $|\eta/k| < 1/2$, and

$$|c| > |k|(|Y| - 1/2).$$

The authentic mask lemma gives, independently of the strong or auxiliary rows,

$$q_0 \geq 16, \qquad q_0(q_0^2 - 1) < R < q_0^4 - 2.$$

Thus $|\ell| > 4q_0^5 > 3R + 1$ in the 118 chart, while $|E| > 4q_0^4 > 3R + 1$ in its companion. These are the early arguments in the retained source 153 Section 2 and the q8 signed source, before any Pell-index interpretation.

Let $d = \ell$ or E. Since $k = R + 1 + hd$ and $|d| > 3R + 1$, either $h = 0$ and $|k| = R + 1$, or $|k| \geq |d| - (R + 1) > 2R \geq R + 1$. Consequently $|c| > R > 0$. For integer T, $b = cT - R$ cannot vanish: $T = 0$ gives $-R$, while $T \neq 0$ gives $|cT| \geq |c| > R$.

Also $|I| \geq 4c^2$ and $\Delta \geq 3$, whence

$$\delta = \Delta^2 I^2 - 1 \geq 144c^4 - 1 > |c| > 0.$$

For every integer y, $t = c + \delta y \neq 0$: this follows from $c \neq 0$ if $y = 0$ and from $|\delta y| > |c|$ otherwise. Therefore $2bt \neq 0$. Equation (3) now gives a rational

$$f = \frac{A_r}{2bt}, \qquad f^2 = F_s \in \mathbb{Z},$$

so f is an integer. Equation (2) gives $E_a = 1$, and $f^2 - \Delta I^2 = 1$ is immediate. Arbitrary signs of every input remain allowed. The cases $c = 0$, $b = 0$ and $t = 0$ were excluded by prior inequalities, not silently divided away.

We have restored a literal zero of the even-strong fifteen-integer kernel identified by equation (2.5) of the retained Round 22 proof. Its Sections 3–4 supply the complete signed bootstrap and *fresh* odd-scale strong and auxiliary witnesses before invoking the authenticated native/compiler/ refutation theorem. We do not incorrectly reuse an even-scale trace in that odd-scale theorem.

1.5 Reverse map and constructive completeness

From any zero of the even-strong kernel, $f^2 = F_s$ and $E_a = 1$ imply $H_r = 0$ without division. Its carrier gives $(Z_{\mathrm{old}}/U + 2L + k)^2 = 4D_\tau$. The rational-root argument makes this quantity an even integer, say 2τ. Delete $f, p_{1,\mathrm{old}}$ and set

$$n = n_{\mathrm{old}} - \sigma L + (2L + k) + p_{1,\mathrm{old}} - \tau.$$

Then $\rho = p_{1,\mathrm{old}} + \tau$, $G = p_{1,\mathrm{old}}^2$, and $D_\tau = \tau^2$, so $N = 0$ and $P = 0$. This works on every old zero with all root signs; no inverse polynomial automorphism is claimed. The retained fresh-witness construction and native completeness theorem therefore give a new zero from every finite syntactic ZFC refutation. Together the two maps establish the ordinary ZFC-provable equivalence of integer unsolvability and syntactic Con(ZFC).

1.6 Complete exact leading forms and verification

Write

$$q_0 = 6\beta j, \quad X_0 = 12q_0w, \quad a_0 = X_0v, \quad F_0 = 12wC - q_0z,$$
$$R_0 = -q_0^3F_0, \quad \ell_0 = X_0v^2, \quad U_0 = 8q_0^3a_0.$$

Identity (1) gives $\deg A_r = 22$, not 38. The whole resultant top is

$$[H_r]_{60} = -2^{20} a_0^{10} r^6 c^{12} R_0^2 y^2.$$

For degree 120,

$$[N]_{60} = U_0^4 R_0^4 (q_0^8 + \ell_0^2)^2.$$

For degree 118 put $k_0 = R_0 + h\ell_0$ and $T_0 = A_c^2 v^2 F_0^2 + 4q_0^2 v^4 + q_0^6$. The exact cancellation

$$(Z_0 + UL)^2 - U^2 D_\tau = Z_0^2 + 2ULZ_0 - U^2(1 + Lk)$$

has degree 26, and gives

$$[N]_{58} = 4U_0^4 \ell_0^2 k_0^4 T_0.$$

The full output tops are the products of these forms. Their 85 and 21 monomials are all expanded by the independent checker. In particular,

$$[j^{58} w^{16} v^{18} z^6 r^6 c^{12} y^2] P_{118} = -2^{124} 3^{74} \beta^{58} \neq 0,$$
$$[j^{66} w^{14} v^{14} z^6 r^6 c^{12} y^2] P_{120} = -2^{126} 3^{80} \beta^{66} \neq 0.$$

The exponents sum to 118 and 120. These are coefficients at the actual fixed numeral slice, not a generic-coefficient estimate or sampled ray.

Required replay uses only Python's standard library. The independent checker imports no producer or circuit implementation, binds every live gate to a separately written formula, checks all signed identities and leading forms, authenticates 255 inherited files and all 615,883 raw coefficient bytes, and rejects both alternative binary operators at each live gate (228 and 224 mutations). Normal and optimized runs have identical explicit acceptance conditions. The separate finite recipe reconstructs all six numeral payloads. Finite checks verify literal artifacts and algebra; the unbounded guarded extraction and inherited logical calibration remain the stated ordinary proofs.